



VP AIR TEST NEET (2025-26)

PART TEST - 01

DROPPER

DURATION: 180 Minutes

Date: 24-08-2025

M. MARKS: 720

Topics Covered

Physics:	Units and Measurements, Mathematical Tools
Chemistry:	Some Basic Concepts of Chemistry, Redox Reactions
Botany:	Cell - The Unit of Life
Zoology:	Structural Organisation in Animals

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. In a hypothetical system of units, five constants are chosen as the fundamental quantities: the speed of light (c), the gravitational constant (G), the Planck constant (h), the Boltzmann constant (k), and the elementary charge (e). What is the dimensional formula for the magnetic permeability of free space (μ_0) when expressed in terms of these five fundamental constants?

- (1) $[c^{-1} G^1 h^0 k^1 e^{-2}]$
- (2) $[c^1 G^0 h^{-1} k^0 e^2]$
- (3) $[c^{-1} G^0 h^1 k^0 e^{-2}]$
- (4) $[c^0 G^1 h^1 k^{-1} e^{-2}]$

2. The relation between few physical quantities is described by the equation

$$P = Q_0 e^{(-\alpha t)} + \frac{\beta}{\alpha} (1 - e^{(-\alpha t)}), \text{ where } t \text{ represents}$$

time. What is the dimension of the quantity β ? ($[P]$ is dimension of quantity P)

- (1) The same as P and Q_0 .
- (2) $[P][T^{-1}]$
- (3) $[P][T]$
- (4) Dimensionless.

3. Match each mathematical operation given in **List-I** with the **correct** number of significant figures that should be in the final answer, given in **List-II**.

List-I (Mathematical Operation)		List-II (Significant Figures in Result)	
A.	$10.0001 - 9.999$	I.	2
B.	$(5.5 \times 10^{-4}) + (2.22 \times 10^{-6})$	II.	4
C.	$(2.000)^3$	III.	1
D.	3.00×1.50	IV.	3

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-I, C-II, D-IV
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-II, C-IV, D-I

4. What is the exact value of the following finite sum?

$$S = \log_{10} \left(1 - \frac{1}{2^2} \right) + \log_{10} \left(1 - \frac{1}{3^2} \right) + \log_{10} \left(1 - \frac{1}{4^2} \right) + \dots + \log_{10} \left(1 - \frac{1}{20^2} \right)$$

- (1) $\log_{10}(21) - \log_{10}(40)$
- (2) $\log_{10}(399) - 3$
- (3) -1
- (4) $\log_{10}(1/2)$

5. An inverted conical tank with a height of 10 meters and a top radius of 5 meters is initially full of water. The water drains from a hole at the vertex such that the rate of change of its volume V is given by:

$$\frac{dV}{dt} = -\frac{\pi}{4} \sqrt{h} \text{ m}^3/\text{s}, \text{ where } h \text{ is the current water}$$

height in meters. At the exact moment when the water level is 4 meters high, at what rate is the water's circular surface area A decreasing?

- (1) $\frac{\pi}{4} \text{ m}^2/\text{s}$
- (2) $\pi \text{ m}^2/\text{s}$
- (3) $\frac{\pi}{8} \text{ m}^2/\text{s}$
- (4) $\frac{1}{8} \text{ m}^2/\text{s}$

6. A student needs to measure the diameter of a thin rod. The professor informs them that the true diameter of the rod is exactly 2.321 cm. The student has access to the four instruments listed below. Which instrument should the student choose to accurately verify this measurement?

- (1) Instrument A (Vernier Caliper): 1 Main Scale Division (MSD) = 1 mm, and 20 Vernier Scale Divisions (VSD) coincide with 19 MSD.
- (2) Instrument B (Vernier Caliper): 1 Main Scale Division (MSD) = 0.5 mm, and 25 Vernier Scale Divisions (VSD) coincide with 24 MSD.
- (3) Instrument C (Screw Gauge): The pitch of the screw is 0.5 mm and the circular head scale has 50 divisions.
- (4) Instrument D (Screw Gauge): The pitch of the screw is 1 mm and the circular head scale has 50 divisions.

7. What is the exact value of the product $\cos(\pi/7) \times \cos(2\pi/7) \times \cos(4\pi/7)$?

- (1) $\frac{1}{8}$
- (2) $-\frac{1}{4}$
- (3) $-\frac{1}{8}$
- (4) $\frac{1}{4}$

8. We can reduce random errors in taking reading from a measuring instrument by:

- (1) Taking large number of observations
- (2) Correcting zero error in instrument
- (3) taking small number of observations
- (4) Both (1) & (2)

9. $\frac{d}{dx}\left(x + \frac{1}{x} + \ln x + \tan x\right)$

(1) $1 - \frac{1}{x^2} + \sec^2 x$

(2) $1 + \frac{1}{x} + \sec^2 x$

(3) $1 + \frac{1}{x^2} + \frac{1}{x} + \sec^2 x$

(4) $1 - \frac{1}{x^2} + \frac{1}{x} + \sec^2 x$

10. The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. The rate at which the area increases, when the side is 10 cm is:

(1) $\sqrt{3} \text{ cm}^2/\text{sec}$

(2) $10 \text{ cm}^2/\text{sec}$

(3) $10\sqrt{3} \text{ cm}^2/\text{sec}$

(4) $\frac{10}{\sqrt{3}} \text{ cm}^2/\text{sec}$

11. A graphic designer is creating a custom logo for a new company. The central element of the logo is a perfect ellipse. On the design software's coordinate grid, the boundary of this ellipse is defined by the following equation, with units in centimeters:

$25x^2 + 16y^2 = 400$. To calculate the amount of ink needed for a solid fill, the designer must find the total area of this elliptical element. What is the area of the ellipse?

(1) $9\pi \text{ cm}^2$ (2) $20\pi \text{ cm}^2$

(3) $25\pi \text{ cm}^2$ (4) $400\pi \text{ cm}^2$

12. In a theoretical model, the thermal conductivity of a rarefied gas under certain extreme conditions is believed to depend on four quantities: the Boltzmann constant (k_B), the gas pressure (P), the mass of a single gas particle (m), and the mean free path of the particles (λ). Using dimensional analysis, which of the following expressions for thermal conductivity is dimensionally correct? (C is a dimensionless constant)

(1) $(Ck_B) \times \frac{Pm}{\lambda}$ (2) $(CP\lambda) \times \sqrt{\frac{k_B}{m}}$

(3) $(Ck_B) \times \sqrt{\frac{P\lambda}{m}}$ (4) $(Ck_B) \times \sqrt{\frac{P}{m\lambda}}$

13. Match trigonometric ratios given in **List-I** with their values given in **List-II** to select the **correct** option.

List-I		List-II	
(A)	$\sin 37^\circ$	(I)	$-\frac{3}{5}$
(B)	$\cos 127^\circ$	(II)	$\frac{3}{5}$
(C)	$\tan 307^\circ$	(III)	$-\frac{4}{3}$
(D)	$\cos 323^\circ$	(IV)	$\frac{4}{5}$

(1) A-II, B-I, C-III, D-IV

(2) A-II, B-IV, C-II, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-III, C-II, D-IV

14. A physics student is studying wave propagation and is checking a list of formulas for the speed (v) of a disturbance in different physical scenarios. The student knows that the speed must depend on the properties of the medium. The list contains formulas that involve the following quantities: Modulus of Elasticity (E), Density (ρ), Surface Tension (S), Coefficient of Viscosity (η), Boltzmann Constant (k_B), Absolute Temperature (T) and Mass of a particle (m). Which of the following formulas for speed (v) is dimensionally **incorrect** and must be an error in the student's notes?

(1) $v = \sqrt{\frac{k_B T}{m}}$

(2) $v = \frac{S}{\eta}$

(3) $v = \frac{E}{\rho}$

(4) Both (2) and (3)

15. Match the physical quantities in **List-I** with the physical quantities in **List-II** that have the same dimensional formula.

List-I		List-II	
A.	Pressure Gradient	I.	Heat Capacity
B.	Latent Heat	II.	Angular Momentum
C.	Planck's Constant	III.	Force Density (Force per unit volume)
D.	Boltzmann Constant	IV.	Gravitational Potential

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-IV, C-II, D-I

(4) A-I, B-III, C-II, D-IV

16. **Statement I:** Dimensionally incorrect equation can never be correct.

Statement II: Physically correct equations must be dimensionally correct.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

17. The value of two resistors are $R_1 = (5 \pm 0.1) \text{ k}\Omega$ and $R_2 = (20 \pm 0.4) \text{ k}\Omega$. The percentage error in the equivalent resistance when they are connected in parallel is:

- (1) 5.125% (2) 2%
- (3) 10.125% (4) 7%

18. The length of a uniform rod is 100.0 cm and radius is 1.00 cm. If length is measured with a meter scale having least count 1 mm and radius is measured with vernier caliper having least count 0.1 mm, the percentage error in calculated volume of cylinder is:

- (1) 2.1% (2) 3%
- (3) 2.5% (4) 3.2%

19. An engineer is designing a decorative support structure for a modern building. The structure's cross-section is defined by the region enclosed between a parabolic support arch and a horizontal beam.

The parabolic arch is described by: $y = x^2$ and

The horizontal beam is described by: $y = 4$

If x and y are measured in meters, what is the total area of this enclosed cross-section?

- (1) $\frac{16}{3} \text{ m}^2$ (2) $\frac{28}{3} \text{ m}^2$
- (3) 12 m^2 (4) $\frac{32}{3} \text{ m}^2$

20. A particle is moving in a circle according to relation $\theta = \left(\frac{3t^3}{10} - 2t \right)$, where θ is in radian and t

is in second. The value of $\frac{d^2\theta}{dt^2}$ at $t = 2 \text{ s}$ is:

- (1) 1.6 rad/s^2 (2) 3.6 rad/s^2
- (3) 2.0 rad/s^2 (4) 1.8 rad/s^2

21. The greatest value of the function $8 \sin \theta - 6 \cos \theta$ is:

- (1) 10 (2) 12
- (3) 20 (4) 15

22. What is the value of the integral

$$\int_{-\pi/2}^{\pi/2} \left[\sqrt{4 - (2\sin\theta)^2} \times 2\cos\theta \right] d\theta$$

- (1) 2π (2) 3π
- (3) 4π (4) π

23. A grandfather clock uses a long brass pendulum to keep time. The period T of the pendulum is

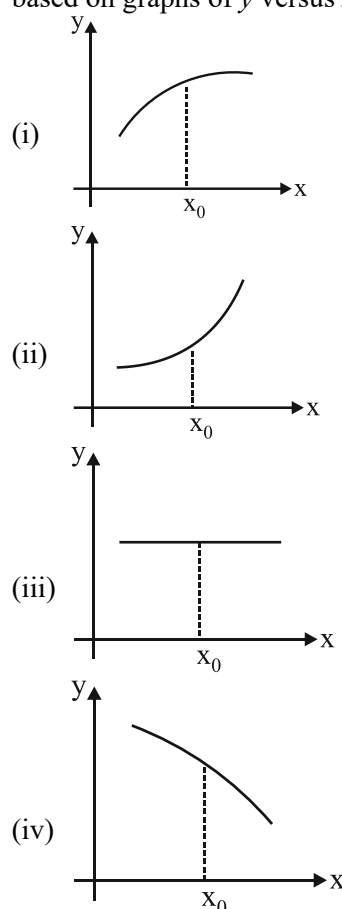
$$T = 2\pi \sqrt{\frac{L}{g}} \text{ where } L \text{ is its length and } g \text{ is}$$

acceleration due to gravity. The clock is perfectly calibrated at a room temperature of 20°C . On a hot summer day, the temperature rises to 30°C . Due to thermal expansion, the length of the brass rod increases according to the formula $L_{\text{final}} = L_{\text{initial}} (1 + \alpha \Delta T)$.

Given that $\alpha = 1.9 \times 10^{-5} / ^\circ\text{C}$, use the binomial approximation to estimate how many seconds the clock loses in one full day.

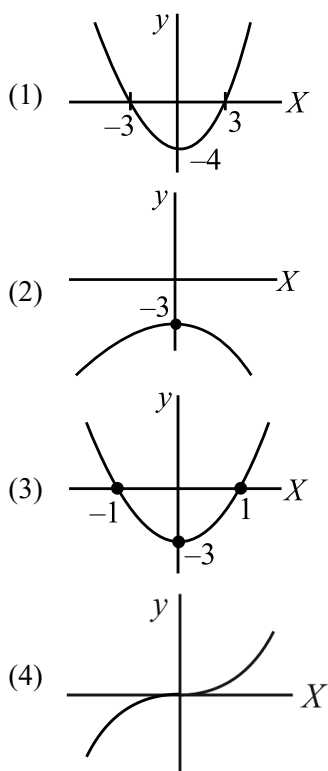
- (1) 1.64 seconds (2) 4.10 seconds
- (3) 8.21 seconds (4) 16.42 seconds

24. Which of the following statement is **incorrect** based on graphs of y versus x as shown below?



- (1) Slope at x_0 is positive and non-zero in graph (i) and (ii).
- (2) Slope is zero in graph (iii).
- (3) Slope at x_0 is negative in graph (iv).
- (4) Slope at x_0 is negative in graph (ii).

25. Most appropriate graph for the equation $y = 3x^2 - 3$ is:



26. A new system of units is proposed in which, unit of mass is α kg, unit of length is β m and unit of time is δ s. What will be the value of 10 J in this new system?

- (1) $10\alpha\beta^2\delta^{-2}$
 (2) $10\alpha^{-1}\beta^{-2}\delta^2$
 (3) $10\alpha^{-2}\beta^{-1}\delta^{-2}$
 (4) $10\alpha^{-1}\beta^2\delta^2$

27. The state of a dynamic system is described by three variables: x, y , and z , linked by:

$$x^2y = \ln(y) + \sin(x)$$

$$y = e^z$$

What is $\frac{dx}{dz}$ at the instant when $z = 0$?

- (1) 1 (2) -1
 (3) 0 (4) e

28. What is the value of the following indefinite integral? (C is constant of integration)

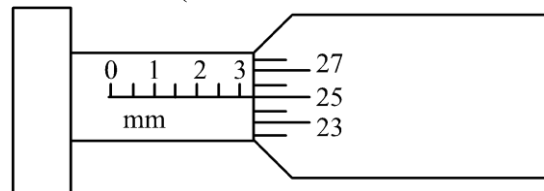
$$\int \left[5\sec^2(x) + (x-2)^2 - (x^2 - 4x + 5\tan^2(x)) \right] dx$$

- (1) $9x + C$
 (2) $x + C$
 (3) $5\tan(x) - x^3 + C$
 (4) $9\tan(x) - x^2 + C$

29. Which of the following formulae for kinetic energy (K) can be ruled out on the basis of dimensional correctness only (where m is mass, v is speed, a is acceleration, $\frac{3}{16}$ is a dimensionless constant)

- (1) $K = m^2a$ (2) $K = \frac{3}{16}mv^2$
 (3) $K = m^2v^3$ (4) Both (1) and (3)

30. The least count of screw gauge shown in figure is 0.01 mm. The value of reading shown on instrument is: (assume no zero error in instrument)



- (1) 3.25 mm (2) 3.24 mm
 (3) 3.30 mm (4) 3.26 mm

31. The 10th term calculated from the end in the AP series 5, 8, 11, ... is 95 then the number of terms in the AP are:

- (1) 38 (2) 40
 (3) 42 (4) 43

32. **Assertion (A):** If two physical quantities have same dimensions, then they can be certainly added or subtracted.

Reason (R): If the dimensions of two physical quantities are same, then both the physical quantities should be same.

- (1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
 (2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
 (3) Assertion (A) is true and Reason (R) is false.
 (4) Both Assertion (A) and Reason (R) are false.

33. A physicist is developing a model for energy dissipation in a turbulent fluid. The proposed equation for the "dissipation rate" Ψ (defined as energy per unit volume per unit time) is given by:

$$\Psi = \frac{(a \times P)}{(x^2 + c)} + \frac{(b \times v^2)}{x} \times e^{(-\lambda t)} \text{ where: } P \text{ is the}$$

pressure, x is a characteristic length, v is the characteristic velocity, t is the time, e is the base of the natural logarithm, a, b, c , and λ are physical constants. What are the dimensions of the constant b ?

- (1) $[ML^{-3}]$ (2) $[ML^{-1}T^{-1}]$
 (3) $[ML^{-2}T^{-1}]$ (4) $[ML^{-1}T^{-2}]$

34. Repeated observation in an experiment gave the values 2.29, 2.24 and 2.34. The mean absolute error in this experiment.

- (1) $\frac{1}{10}$ (2) $\frac{1}{20}$
(3) $\frac{1}{30}$ (4) $\frac{1}{40}$

35. Approximate value of $(2.004)^3$ using binomial expansion is:

- (1) 8.048 (2) 8.081
(3) 8.024 (4) 8.102

36. The minimum value of $y = 5x^2 - 2x + 5$ is:

- (1) $\frac{22}{5}$ (2) $\frac{26}{5}$
(3) $\frac{24}{5}$ (4) $\frac{27}{5}$

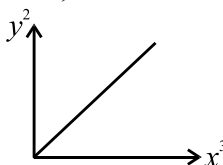
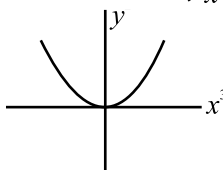
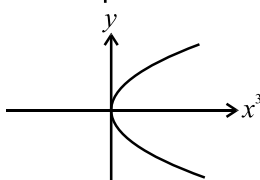
37. A student is tasked with determining the density of a solid sphere. To do this, he performs two measurements: He measures the mass (m) of the sphere using a digital balance, recording the measurement as (40.00 ± 0.02) g. He measures the diameter (d) of the sphere using a screw gauge with a least count of 0.001 cm. The reading is 2.000 cm. What is the maximum percentage error in the calculated value of the density?

- (1) 0.15% (2) 0.17%
(3) 0.05% (4) 0.20%

38. The initial and final temperature of water as recorded by an observer are $(40.6 \pm 0.2)^\circ\text{C}$ and $(78.3 \pm 0.3)^\circ\text{C}$. Calculate the rise in temperature with error limits.

- (1) $(37.7 \pm 0.3)^\circ\text{C}$ (2) $(37.7 \pm 0.2)^\circ\text{C}$
(3) $(37.7 \pm 0.1)^\circ\text{C}$ (4) $(37.7 \pm 0.5)^\circ\text{C}$

39. If $y^2 = 3x^3$, then choose the **correct** graph.

- (1) 
(2) 
(3) 
(4) Both (1) and (3)

40. The order of magnitude of 5×10^7 is:

- (1) 8 (2) 7
(3) 6 (4) 9

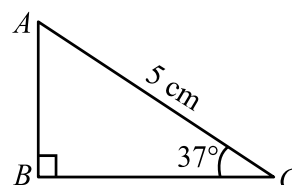
41. There are 100 divisions on the circular scale of a screw gauge of pitch 1 mm. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found the 4 linear scale divisions are clearly visible while 60th division on circular scale coincide with the reference line. The diameter of the wire is:

- (1) 4.65 mm (2) 4.55 mm
(3) 4.60 mm (4) 3.35 mm

42. In the measured value 0.0021020, the number of zeroes that are significant is/are

- (1) 4 (2) 5
(3) 2 (4) 1

43. Find the perimeter of the triangle ABC given below.



- (1) 10 cm (2) 12 cm
(3) 9 cm (4) 14 cm

44. Which of the following options contains the **correct** trigonometric formula?

- A. $\cos(180^\circ - \theta) = -\cos\theta$
B. $\cos(2\theta) = \cos^2\theta - \sin^2\theta$
C. $\tan(A + B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

- D. $\sin(-\theta) = \sin\theta$

- (1) A and B only (2) B and C only
(3) C and D only (4) A and D only

45. A theoretical physicist is exploring relationships between five distinct physical quantities from different fields of study. The quantities are: V : (Rate of change of Stress), W : Angular Momentum, X : "Energy Flux Density", Y : Planck's Constant, Z : Power. Which of the following expressions represents a dimensionally possible operation?

- (1) $\frac{WX}{V} + Y$ (2) $e^{(W/Y)}$
(3) $\sin(V + W)$ (4) $VY - Z$

46. The maximum number of molecules is present in:
(1) 16 g of O_2
(2) 1 mole of N_2
(3) 5 g of H_2
(4) 11.2 L of CO_2 at STP
47. In $[Cr(O_2)(NH_3)_4H_2O]Cl_2$, oxidation number of Cr is +3, then oxygen will be in the form:
(1) dioxide
(2) peroxide
(3) superoxide
(4) oxide
48. 9.5 g of hydrocarbon on combustion with 32.8 g of oxygen produces 24.7 g of CO_2 and 17.6 g of H_2O . This data illustrates:
(1) Avogadro's Law
(2) Law of Conservation of Mass
(3) Law of Definite Proportions
(4) Law of Multiple Proportions
49. Which of the following has 3 significant figures?
(1) 83.000
(2) 0.0830
(3) 0.008300
(4) 8300
50. Magnesium reacts with HCl producing H_2 and corresponding magnesium salts. In such reactions, Mg undergoes:
(1) reduction only
(2) oxidation only
(3) neither oxidation nor reduction
(4) both oxidation and reduction
51. Amount of oxygen required to burn completely 129 g of hexane is:
(1) 456 g
(2) 798 g
(3) 532 g
(4) 312 g
52. The standard reduction potentials at 25 °C of Li^+/Li , Ba^{2+}/Ba , Na^+/Na and Mg^{2+}/Mg are -3.05, -2.73, -2.71 and -2.37 volt respectively. Which one of the following is the strongest oxidising agent?
(1) Na^+
(2) Li^+
(3) Ba^{2+}
(4) Mg^{2+}
53. The compound AB completely decomposes into A and B on heating. 100 g of AB, on strong heating, gave 75 g of A. How much quantity of AB should be decomposed by heating to obtain 20 g of B?
(1) 45 g
(2) 80 g
(3) 40 g
(4) 25 g
54. For the redox reaction,
 $HgCl_2 + SnCl_2 \rightarrow 2Hg + SnCl_4$,
the **correct** coefficients of $HgCl_2$ and $SnCl_2$ for the balanced equation respectively are:
(1) 1, 1
(2) 1, 2
(3) 2, 2
(4) 2, 1
55. Which of the following concentration term is temperature independent?
A. %w/w
B. Molarity
C. Mole fraction
D. Molality
Choose the **most** appropriate answer from the options given below:
(1) A and B only
(2) A, B and C
(3) A and D only
(4) A, C and D
56. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): The average oxidation state of sulphur in $S_4O_6^{2-}$ ion is +2.5.
Reason (R): The oxidation state of each terminal sulphur in $S_4O_6^{2-}$ ion is +5 while the two middle sulphurs has zero oxidation state.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is false but R is true.
(2) A is true but R is false.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.
57. In the reaction, $8Al + 3Fe_3O_4 \rightarrow 4Al_2O_3 + 9Fe$, the number of electrons transferred from reductant to oxidant is:
(1) 8
(2) 4
(3) 16
(4) 24

58. Given below are two statements:
Statement I: The oxidation state of H in LiH is -1 .
Statement II: The oxidation state of P in P_4 is zero.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) Statement I and Statement II both are correct.
 (2) Statement I and Statement II both are incorrect.
 (3) Statement I is correct, but Statement II is incorrect.
 (4) Statement I is incorrect, but Statement II is correct.
59. For the reaction,
 $CaO + 2HCl \rightarrow CaCl_2 + H_2O$
 14 g of CaO is reacted with excess of hydrochloric acid and 13.875 g of $CaCl_2$ is formed. The percent yield of $CaCl_2$ is:
 (1) 25%
 (2) 100%
 (3) 50%
 (4) 75%
60. An organic compound has a molecular weight of 200 g. If it contains 42% by mass of nitrogen, then the number of atoms of nitrogen in one molecule of the organic compound is:
 (1) 6
 (2) 3
 (3) 4
 (4) 5
61. The oxidation state of C in $C_6H_{12}O_6$ is equal to the oxidation state of C in:
 (1) $HCOOH$
 (2) $HCHO$
 (3) CH_4
 (4) CO
62. Chlorine occurs in two isotopic forms, ^{35}Cl and ^{37}Cl . If the average atomic mass of chlorine is 35.5, then the ratio of the percentage abundance of ^{37}Cl to ^{35}Cl is:
 (1) 1 : 3
 (2) 2 : 1
 (3) 1 : 1
 (4) 3 : 1
63. The oxidation states of sulphur in Marshall's acid is:
 (1) +6
 (2) +4
 (3) -6
 (4) -4
64. If the density of water at room temperature is 1 g/mL, then the number of molecules in one drop of water having volume 9 mL is:
 (1) 12.044×10^{23}
 (2) 12.044×10^{22}
 (3) 3.011×10^{23}
 (4) 3.011×10^{22}
65. Which among the following is **not** a mixed oxide?
 (1) Fe_3O_4
 (2) Pb_3O_4
 (3) Al_2O_3
 (4) Mn_3O_4
66. 21 g of N_3^- ion [Molar mass = 42 g mol^{-1}] contains:
 A. $11 N_A$ electrons
 B. $11 N_A$ protons
 C. $10.5 N_A$ electrons
 D. $10.5 N_A$ protons
 Choose the **most** appropriate answer from the options given below:
 (1) A and B
 (2) B and C
 (3) C and D
 (4) A and D
67. The equivalent weight of ClO_3^- acting as reducing agent is:
 (1) $\frac{M}{4}$
 (2) $\frac{M}{2}$
 (3) $\frac{M}{7}$
 (4) M
68. Number of moles present in 1 m^3 of any gas at STP is:
 (1) 32.61
 (2) 89.28
 (3) 44.64
 (4) 64.33
69. The vapour density of a mixture containing NO_2 and N_2O_4 is 34.5 at 27°C . The mass of NO_2 in 100 g of the mixture is:
 (1) 66.66 g
 (2) 82.61 g
 (3) 72.93 g
 (4) 33.33 g

70. Which of the following is **not** a disproportionation reaction?

- (1) $\text{P}_4 + 3\text{OH}^- + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$
- (2) $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{ClO}^- + \text{Cl}^- + \text{H}_2\text{O}$
- (3) $2\text{F}_2 + 2\text{OH}^- \rightarrow 2\text{F}^- + \text{OF}_2 + \text{H}_2\text{O}$
- (4) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$

71. Match List I with List II

List I (Multiple)		List II (Prefix used in SI system)	
(A)	10^{-21}	(I)	zepto
(B)	10^{12}	(II)	tera
(C)	10^{-12}	(III)	zeta
(D)	10^{21}	(IV)	pico

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-III, C-II, D-IV
- (4) A-II, B-I, C-IV, D-III

72. The oxidation state of oxygen is -1 in;

- (1) ClO_2^-
- (2) H_2O_2
- (3) KO_2
- (4) RbO_2

73. Given below are two statements:

Statement I: 30.11×10^{23} neutrons are present in 9 mL of water.

Statement II: Volume at STP occupied by 7 g of CO gas is 5600 mL.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

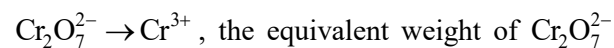
- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

74. A container contains 280 g of N_2 gas, out of which 6.022×10^{22} molecules of it are removed. How many moles of N_2 are left in container?

- (1) 0.99 moles
- (2) 10 moles
- (3) 9.9 moles
- (4) 0.1 moles

75. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): In the given half reaction,



[Molar mass = M] is $\frac{M}{3}$.

Reason (R): The change in oxidation state per chromium atom is three.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

76. If an organic compound containing C and H has 96% of carbon, then the empirical formula of the compound will be:

- (1) C_2H
- (2) C_4H
- (3) CH_2
- (4) CH_4

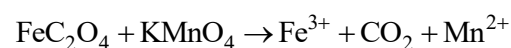
77. "Equal volumes of all gases at the same temperature and pressure should contain equal number of molecules." This statement corresponds to:

- (1) Law of Definite proportions
- (2) Law of Multiple proportions
- (3) Avogadro's Law
- (4) Gay Lussac's Law of gaseous volumes

78. Which of the following reaction is **not** a redox reaction?

- (1) $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(s)} + \text{H}_2\text{(g)}$
- (2) $\text{HCl(g)} + \text{NaOH(s)} \rightarrow \text{NaCl(s)} + \text{H}_2\text{O(l)}$
- (3) $\text{TiCl}_4\text{(l)} + 2\text{Mg(s)} \rightarrow \text{Ti(s)} + 2\text{MgCl}_2\text{(s)}$
- (4) $3\text{Mg(s)} + \text{N}_2\text{(g)} \rightarrow \text{Mg}_3\text{N}_2\text{(s)}$

79. The species which are oxidised and reduced in the given reaction are:



- (1) Oxidised : Fe, C ; Reduced : Mn
- (2) Oxidised : Mn ; Reduced : Fe, C
- (3) Oxidised : Mn, C ; Reduced : Fe
- (4) Oxidised : Fe ; Reduced : Mn, C

80. If 20 g of MCl_3 contains 0.8 mol of chlorine atoms then, molecular mass of MCl_3 is:
- (1) 91 g/mol
 - (2) 25 g/mol
 - (3) 82 g/mol
 - (4) 75 g/mol
81. The number of moles of MnO_4^- required to oxidise 1 mole of H_2O_2 into O_2 in acidic medium is:
- (1) $\frac{2}{5}$
 - (2) $\frac{5}{2}$
 - (3) 2
 - (4) 5
82. Which of the following compounds can **not** undergo disproportionation reaction?
- (1) SO_2
 - (2) H_2O_2
 - (3) MnO_2
 - (4) HClO_4
83. 30 g of C_2H_6 is burnt in presence of excess oxygen. The maximum moles of CO_2 formed is:
- (1) 1.5 mol
 - (2) 2 mol
 - (3) 1 mol
 - (4) 2.5 mol
84. Which among the following is the strongest oxidising agent in standard state?
- (1) F_2
 - (2) O_2
 - (3) H^+
 - (4) Li^+
85. The ratio of molecular masses of two gases having vapour density in the ratio 1 : 2 will be:
- (1) 2:1
 - (2) 1:2
 - (3) 4:1
 - (4) 4:4

86. Match List I with List II

List I (Reaction)		List II (Type of reaction)	
A.	$\text{C(s)} + \text{O}_2(\text{g}) \xrightarrow{\Delta} \text{CO}_2(\text{g})$	I.	Non-metal displacement reaction
B.	$2\text{H}_2\text{O(l)} \xrightarrow{\Delta} 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$	II.	Combination reaction
C.	$\text{CuSO}_4(\text{aq}) + \text{Zn(s)} \rightarrow \text{Cu(s)} + \text{ZnSO}_4(\text{aq})$	III.	Decomposition reaction
D.	$2\text{H}_2\text{O(l)} + 2\text{F}_2(\text{g}) \rightarrow 4\text{HF(aq)} + \text{O}_2(\text{g})$	IV.	Metal displacement reaction

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
 - (2) A-II, B-III, C-I, D-IV
 - (3) A-III, B-II, C-IV, D-I
 - (4) A-III, B-II, C-I, D-IV
87. The mass of SO_2 [Molar mass = 64 g mol^{-1}] produced by the reaction of 38 g of CS_2 [Molar mass = 76 g mol^{-1}] and 64 g of O_2 [Molar mass = 32 g mol^{-1}] is 16 g. The percent yield of the reaction is:
- $$\text{CS}_2 + 3\text{O}_2 \rightarrow 2\text{SO}_2 + \text{CO}_2$$
- (1) 50%
 - (2) 70%
 - (3) 25%
 - (4) 65%
88. For the given reaction, which of the following statement is **correct**?
- $$\text{Sn}^{2+} + 2\text{Fe}^{3+} \rightarrow 2\text{Fe}^{2+} + \text{Sn}^{4+}$$
- (1) Sn^{2+} is getting oxidised
 - (2) Sn^{4+} is getting oxidised
 - (3) Fe^{2+} is getting reduced
 - (4) This is not a redox reaction
89. The equivalent weight of MnO_4^- [Molar mass=M] when it is converted to Mn^{2+} ion is;
- (1) $\frac{M}{5}$
 - (2) $\frac{M}{3}$
 - (3) $\frac{M}{7}$
 - (4) M
90. The **correct** decreasing order of oxidation state of 'Mn' in the given compounds is:
- (I) K_2MnO_4
 - (II) KMnO_4
 - (III) MnO_2
 - (IV) Mn_2O_3
- (1) I > II > III > IV
 - (2) II > I > III > IV
 - (3) I > II > IV > III
 - (4) II > I > IV > III

91. Identify the correct statements from the following.
- The outer membrane of the nuclear envelope usually remains continuous with the endoplasmic reticulum.
 - Nucleoli are membrane-bound structures within the nucleoplasm.
 - The nuclear pores are passages for the movement of RNA and protein molecules in one direction only, from the nucleus to the cytoplasm.
 - The nucleolus is a primary site for active ribosomal RNA synthesis.
 - The interphase nucleus has a loose and indistinct network of nucleoprotein fibres.

Choose the **correct** answer from the options below:

- (1) A, D, E (2) A, B, C
- (3) B, C, D (4) C, D, E

92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The smooth endoplasmic reticulum is the major site for synthesis of lipid.

Reason (R): In animal cells, lipid-like steroidal hormones are synthesised in RER.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

93. Match List-I with List-II.

List-I		List-II	
(A)	Rudolf Virchow	(I)	Discovered the nucleus
(B)	Theodore Schwann	(II)	First saw and described a live cell
(C)	Robert Brown	(III)	All cells arise from pre-existing cells
(D)	Antonie Von Leeuwenhoek	(IV)	Presence of cell wall is a unique character of plant cells

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

94. Choose the correct statement.

- (1) A polysome synthesizes sugars in prokaryotes.
- (2) A polysome is made up of many ribosomes attached to a single mRNA.
- (3) A polysome is a polymer of ribosomes only.
- (4) A polysome is made up of RNA, DNA and ribosomes.

95. If the middle lamella between plant cells were to be dissolved, what would be the most direct consequence?

- (1) The cells would lose their turgidity and shrink.
- (2) Transport of substances between cells would stop.
- (3) The cells would separate from each other.
- (4) The plasma membranes of the cells would rupture.

96. Arrange the following cells/organisms in ascending order of their size.

- Mycoplasma
- Ostrich egg
- Typical bacteria
- Human RBCs
- Typical eukaryotic cell

Choose the correct sequence:

- (1) $A < C < E < D < B$
- (2) $A < D < C < E < B$
- (3) $C < A < D < E < B$
- (4) $A < C < D < E < B$

97. Given below are two statements:

Statement I: The eukaryotic ribosomes are 80S, where 'S' is Svedberg's unit and is the direct measure of density and size.

Statement II: The two subunits of 80S ribosomes are 50S and 30S.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

98. Select the **incorrectly** matched pair.
- (1) Pili – Motility in bacteria
 - (2) Fimbriae – Attachment to host tissues
 - (3) Flagella – Bacterial motility
 - (4) Cilia – Never present in bacteria
99. Which of the given statements are true for the cell envelope of a prokaryotic cell?
- A. It consists of a tightly bound five-layered structure.
 - B. The layers act together as a single protective unit.
 - C. The glycocalyx could be a loose sheath called capsule or a thick layer called slime layer.
 - D. The cell wall prevents the bacterium from bursting or collapsing.
 - E. All prokaryotes have a cell wall as part of their cell envelope.
- The correct option is:
- (1) A, B, D
 - (2) B, D only
 - (3) C, E only
 - (4) B, D, E

100. Choose the correct statement.
- (1) The central proteinaceous part of the proximal region of the centriole is called the axoneme.
 - (2) Each centrosome has two centrioles that are surrounded by amorphous pericentriolar materials.
 - (3) The centrosome present inside the centriole has a cartwheel like structure.
 - (4) The hub of a centriole is connected with tubules of the peripheral triplets by radial spokes made of lipids.

101. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Lysosomes	(I)	Site of aerobic respiration
(B)	Mitochondria	(II)	Digestion of macromolecules
(C)	Golgi apparatus	(III)	Stores materials not useful for cell
(D)	Vacuole	(IV)	Formation of glycoproteins

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

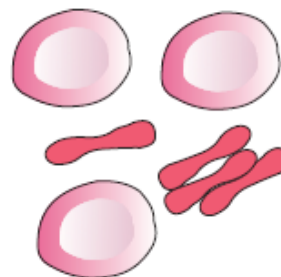
Assertion (A): The inner membrane of mitochondria forms a number of infoldings called the thylakoids.

Reason (R): The infoldings of inner mitochondrial membrane increase the surface area for ATP-generating chemical reactions.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

103. Shown below is a type of human cell. Choose the incorrect statement about it.



- (1) It does not contain nucleus at maturity.
- (2) The plasma membrane contains 40 per cent lipids.
- (3) The plasma membrane contains 52 per cent proteins.
- (4) The cell wall is made of calcium pectate.

104. Read the following statements about the fluid mosaic model of plasma membrane.

- A. It was proposed by Singer and Nicolson.
- B. It describes the cell membrane as a protein bilayer with associated lipids.
- C. According to this model, lipids can undergo lateral movement, but proteins cannot.
- D. The fluidity of the membrane is important for functions like cell division and secretion.
- E. The model applies only to animal cell membranes.

The correct statements are;

- (1) A and D only
- (2) B and C only
- (3) A, D and E
- (4) A, C and D

105. The pigments that give the part of the plant a yellow, orange or red colour are present in;

- (1) chromatophores.
- (2) chromoplasts.
- (3) chloroplasts.
- (4) leucoplasts.

106. Given below are two statements:

Statement I: The cell wall is a rigid and living structure that not only gives shape to the cell but also protects it from mechanical damage and infection.

Statement II: The cell wall of a young plant cell, the secondary wall, is capable of growth.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

107. Match List-I with List-II.

List-I (Chromosome type)		List-II (Characteristic)	
(A)	Metacentric	(I)	Centromere is close to one end
(B)	Sub-metacentric	(II)	Terminal centromere
(C)	Acrocentric	(III)	Middle centromere forming two equal arms
(D)	Telocentric	(IV)	Centromere slightly away from the middle

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

108. An animal cell is in urgent requirement of an ion. This cell although has adequate concentration of this ion inside, yet it needs to transport it from the surrounding fluid. What kind of transport will be required for the transport of this ion into the cell?

- (1) Osmosis involving movement of substance along the concentration gradient
- (2) Simple diffusion involving movement of substance with water
- (3) Passive transport that requires transporter proteins
- (4) Active transport that requires energy and carrier proteins

109. Select the option having correct set of organelles that are included in the endomembrane system.

- A. Endoplasmic reticulum
- B. Mitochondria
- C. Golgi complex
- D. Lysosomes
- E. Vacuoles
- F. Peroxisomes

- (1) A, B, C, D
- (2) A, C, D, E
- (3) B, C, E, F
- (4) A, C, D, F

110. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The Golgi apparatus remains in close association with the endoplasmic reticulum.

Reason (R): Materials packaged as vesicles from the ER fuse with the *trans* face of the Golgi apparatus.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

111. Choose the incorrect statement for the cell organelle shown below.



- (1) Present in both prokaryotic and eukaryotic cells.
- (2) Present on the surface of smooth endoplasmic reticulum.
- (3) Present in matrix of mitochondria.
- (4) Present in stroma of chloroplasts.

112. Match List-I with List-II.

List-I		List-II	
(A)	Plasmodesmata	(I)	Component of cell wall in algae
(B)	Phosphate granules	(II)	Finger-like projections in animal cells
(C)	Microvilli	(III)	Store reserve material in prokaryotes
(D)	Mannans	(IV)	Traverse the middle lamella and cell wall in plant cells

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

113. Identify the true statement about mesosome.

- (1) It is an infolding of cell wall.
- (2) It confers antibiotic resistance in bacteria.
- (3) It helps in replication of genetic material.
- (4) It helps in formation of plasma membrane.

114. Arrange the following steps in the correct sequence for the synthesis and secretion of a protein.

- A. Movement towards the maturing face of Golgi cisternae.
- B. Fusion of vesicle from ER with the forming face of Golgi cisternae.
- C. Modification of protein in Golgi cisternae.
- D. Synthesis of protein on RER.
- E. Release from the *trans* face of Golgi cisternae.

Choose the correct option:

- (1) D - B - A - C - E
- (2) D - C - B - A - E
- (3) D - B - C - A - E
- (4) B - D - C - E - A

115. Given below are two statements:

Statement I: The stroma lamellae connect the lumens of the different thylakoids.

Statement II: The axoneme of a flagellum consists of 20 microtubules arranged in a 9+2 array.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

116. Which of the following statements about mitochondria is/are incorrect?

- A. The components required for protein synthesis are present between outer and inner membrane.
- B. The number of mitochondria is constant for all cells.
- C. The matrix contains a single circular DNA molecule and 80S ribosomes.
- D. They divide by fission.
- E. They are easily visible under the microscope without specific staining.

Choose the correct option.

- (1) A, B, C, E (2) D only
- (3) A, B, C only (4) A and E only

117. Choose the correct statement.

- (1) The main arena of cellular activities in both the plant and animal cells is the nucleus.
- (2) In both prokaryotic and eukaryotic cells, a semi-fluid matrix occupies the volume of the cell.
- (3) The plasmids of bacterial DNA are larger than the genomic DNA.
- (4) Various chemical reactions occur in the cell membrane it to keep the cell in the 'living state'.

118. Identify the correctly matched pair.

- (1) Lysosome – Formed by the process of packaging in the ER
- (2) Golgi apparatus – Divides the intracellular space into luminal and extra luminal compartments
- (3) Vacuole – Contains genetic material in plant cells
- (4) Smooth endoplasmic reticulum – Do not bear ribosomes on surface

119. The cell theory emphasized upon the;
- (1) use of cell-free systems to investigate physiological and behavioural processes.
 - (2) identifying the types of organic compounds present in living organisms.
 - (3) unity underlying the diversity of forms.
 - (4) identifying the molecular basis of all physiological processes.

120. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): RER is frequently observed in cells actively involved in protein synthesis and secretion.

Reason (R): Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

121. While observing a plant cell under microscope, it was found that about 85% of its volume was occupied by a single organelle. Choose the correct statement about this organelle.

- (1) It consists of hydrolytic enzymes that are inactive at acidic pH.
- (2) This organelle is double-membrane bound and also present in animal cells.
- (3) Its membrane facilitates the transport of ions present in the cytoplasm into it against the concentration gradient.
- (4) It is a part of cytoskeleton made up of microfilaments.

122. Read the following statements.

- A. Cilia and flagella are hair-like outgrowths of the cell membrane.
- B. Centrosome is a non-membrane bound organelle found in both prokaryotic and eukaryotic cells.
- C. Centrioles are not present in most of the plant cells.

D. Centrioles form spindle fibres during cell division in animal cells.

E. Both cilia and flagella emerge from centriole-like structures called basal bodies.

In the light of above statements choose the correct answer from the options given below:

- (1) A, B, C and E are incorrect
- (2) A, B, C and D are correct
- (3) B, C, D and E are incorrect
- (4) A, C, D and E are correct

123. Match List-I with List-II.

List-I		List-II	
(A)	Perinuclear space	(I)	0.5 μm
(B)	Average diameter of a mitochondrion	(II)	10 - 50 nm
(C)	Diameter of Golgi cisternae	(III)	15 nm by 20 nm
(D)	Size of bacterial ribosome	(IV)	0.5 μm to 1.0 μm

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

124. Given below are two statements:

Statement I: In plants, the lysosome is bound by tonoplast.

Statement II: In *Amoeba*, the contractile vacuole is important for osmoregulation and excretion.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

125. The microbodies;

- (1) are membrane bound structures.
- (2) are large vesicles.
- (3) contain enzymes required for photosynthesis.
- (4) are found in plant cells but not in animal cells.

- 126.** Which of the following statements are correct?
- A. Plastids are found in all plant cells and euglenoids.
 - B. Chloroplasts are bound by a double membrane made of proteins only.
 - C. Leucoplasts are colourless plastids of varied shapes and sizes.
 - D. Amyloplasts store carbohydrates.
 - E. Aleuroplasts store fats and oils.
- Choose the correct option.
- (1) A, C and D
 - (2) B and C only
 - (3) A, D and E
 - (4) A, B and D
- 127.** Identify the correct statement.
- (1) The 'satellite' appears in a chromosome due to the presence of primary constriction.
 - (2) Histone proteins are acidic in nature.
 - (3) Kinetochore holds two chromatids of a chromosome.
 - (4) The length of DNA in a human cell is approximately 2 metres.
- 128.** The genetic material in a prokaryotic cell is:
- (1) enveloped by a nuclear membrane.
 - (2) basically naked.
 - (3) contained within multiple linear chromosomes.
 - (4) associated with non-histone proteins to form chromatin.
- 129.** How many of the following statements are false?
- A. All eukaryotic cells are identical.
 - B. The eukaryotes include all the protists, plants, animals and fungi.
 - C. In eukaryotic cells there is an extensive compartmentalisation of cytoplasm through the presence of membrane bound organelles.
 - D. The genetic material in eukaryotes is organised into chromosomes.
 - E. Eukaryotic cells have a variety of simple locomotory and cytoskeletal structures.
- (1) Four
 - (2) Three
 - (3) Two
 - (4) Five
- 130.** Theodore Schwann was a;
- (1) British zoologist.
 - (2) German botanist.
 - (3) German zoologist.
 - (4) British botanist.

- 131.** Choose the correct statement.
- (1) The proteins in the plasma membrane can be present on the surface as well as embedded within it.
 - (2) The proteins in the plasma membrane have been classified as integral or peripheral based on their three-dimensional structure.
 - (3) The tail of phospholipids in the plasma membrane is polar and hence placed towards the inner part.
 - (4) The head of phospholipids in the plasma membrane is non-polar and is exposed to the external environment.
- 132.** Unicellular organisms are not capable of:
- (1) independent existence.
 - (2) performing the essential functions of life.
 - (3) dividing.
 - (4) forming tissues and organs.
- 133.** Choose the correct statement.
- (1) The inner chloroplast membrane is relatively more permeable than the outer membrane.
 - (2) Being small, plastids cannot be easily visualised under the microscope.
 - (3) Chlorophyll pigments are present in the thylakoids.
 - (4) The ribosomes of the chloroplasts are larger than the cytoplasmic ribosomes.
- 134.** A plant cell is treated with a toxin that specifically and exclusively inactivates 40S ribosomes. Which of the following describes the most immediate effect on protein synthesis within this cell?
- (1) All protein synthesis will cease immediately, as both cytoplasmic and organellar protein synthesis will be halted.
 - (2) The synthesis of enzymes for the cytoplasm will be halted, but protein synthesis within mitochondria will continue.
 - (3) The synthesis of proteins within the chloroplasts will stop, but the synthesis of enzymes in the cytoplasm will continue normally.
 - (4) Only the synthesis of proteins destined for secretion outside the cell will be affected.
- 135.** Different bacteria respond differently to the Gram staining procedure. It is based on differences in which of the following?
- (1) The structure of the pili and fimbriae.
 - (2) The presence or absence of a mesosome.
 - (3) The differences in the cell envelope.
 - (4) The structure of the flagellar basal body.

- 136.** In higher animals, which statement **best** captures how an organ system comes into existence?
- (1) One tissue type executes all body functions for survival.
 - (2) Several distinct organs integrate (physically/chemically) to achieve a shared function.
 - (3) An organ system is simply a cluster of similar cells plus matrix.
 - (4) Each cell independently performs every life activity within an organ system.

- 137.** Which statement **best** describes dense regular connective tissue?
- (1) Fibres random; collagen absent.
 - (2) Fibroblasts between irregular collagen bundles.
 - (3) Parallel collagen bundles with fibroblasts in rows.
 - (4) Attaches skin to muscle.

- 138.** Match **List-I** with **List-II**.

List-I		List-II	
(A)	Tendons	(I)	Tip of nose
(B)	Fibroblasts	(II)	Attach bones to bones
(C)	Hyaline cartilage	(III)	Attach muscles to bones
(D)	Ligaments	(IV)	Fibres secreting cells

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 - (2) A-I, B-II, C-III, D-IV
 - (3) A-III, B-IV, C-II, D-I
 - (4) A-IV, B-II, C-I, D-III
- 139.** Read the following statements and choose the combination with **correct** statements.
- A. Chondrocytes are the matrix producing cells of cartilage.
 - B. Bone is rich in elastin and flexible due to low Ca^{2+} .
 - C. Osteocytes in lacunae maintain bone matrix.
 - D. In adults, cartilage occurs in joints, pinna and intervertebral discs.
- (1) A, B and C
 - (2) A, C and D
 - (3) B and D only
 - (4) A and D only

- 140.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion A: The survival of multicellular organisms is significantly higher than that of unicellular organisms due to the organization of cells into tissues.

Reason R: This organization allows for a division of labor, where specialized cells perform specific functions efficiently, improving the overall function of the organism.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
 - (2) Both A and R are true, but R is not the correct explanation of A.
 - (3) A is true, but R is false.
 - (4) A is false, but R is true.
- 141.** In the areolar tissue, which cell is phagocytic in nature and which one secretes histamine?
- (1) Phagocytic: Fibroblast; Histamine-secreting: Mast cell
 - (2) Phagocytic: Macrophage; Histamine-secreting: Mast cell
 - (3) Phagocytic: Mast cell; Histamine-secreting: Macrophage
 - (4) Phagocytic: Fibroblast; Histamine-secreting: Macrophage
- 142.** Mammalian bone differs from cartilage in the presence of;
- (1) Lymph vessels
 - (2) Collagen
 - (3) Matrix
 - (4) Haversian canals
- 143.** What is the chief supportive role of areolar tissue?
- (1) Connects osteocytes in bone.
 - (2) Forms flexible support beneath epithelia.
 - (3) Forms neuroglial cells
 - (4) Links chondrocytes to cartilage.
- 144.** Given below are two statements:
- Statement I:** Tendons are composed of dense regular connective tissue.
- Statement II:** Ligaments are made of dense irregular connective tissue, allowing for strength in multiple directions.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Statement I is correct but Statement II is incorrect.
 - (3) Statement I is incorrect but Statement II is correct.
 - (4) Both Statement I and Statement II are incorrect.

145. Following a skin wound, a tissue sample from the healing area is examined. The pathologist observes elongated, spindle-shaped cells actively secreting a meshwork of protein fibers that will form the scar tissue. These cells are surrounded by a gel-like ground substance which they also produce. The pathologist is observing:
- (1) Macrophages
 - (2) Fibroblasts
 - (3) Mast cells
 - (4) Plasma cells
146. Choose the **incorrect** option w.r.t types of tissues and their function.
- (1) Dendrite: Receives impulse
 - (2) Axon: Transmits impulse away from cell body
 - (3) Blood cells: All enucleated
 - (4) Muscle: Many, long cylindrical fibres in parallel arrays
147. Identify the **incorrect** statement among the following.
- (1) Nucleus is marginalised in adipocytes.
 - (2) Compound epithelium is multilayered i.e has more than one layer of cells.
 - (3) Pseudostratified tissue is a type of stratified epithelium.
 - (4) The matrix, or ground substance, is a key component that determines the physical properties of the tissue.
148. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Compound epithelium is not found in the stomach lining.
Reason R: Compound epithelium plays major role in absorption, secretion and excretion.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both A and R are true and R is the correct explanation of A.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is false but R is true.
149. Which of the following is **not** true?
- (1) Limb bones are long bones.
 - (2) The function of long bones is weight bearing and bringing out movement.
 - (3) Bone marrows in all long bones are site of blood cells production.
 - (4) Bones protect softer tissues and organs.
150. Which of the following consequences is most likely if the intercalated discs between cardiac muscle cells are destroyed due to disease?
- (1) Skeletal muscles will lose their striations
 - (2) The heart will contract irregularly due to disrupted electrical impulse conduction
 - (3) Smooth muscle contraction will become voluntary
 - (4) Bone tissue will lose mechanical strength
151. How many of the following statements are **correct**?
- A. Neurons and muscle fibres share a similar property of responding to stimulus.
 - B. Neurons are the unit of neural system.
 - C. Neuroglia makes less than 50% of total volume of neural system.
 - D. The stimulation or inhibition of an adjacent neuron depends upon the disturbance arriving at the output zone of previous neuron.
 - E. Neural tissue is primarily responsible for an organism to respond to changes in their environment.
- (1) Three
 - (2) Four
 - (3) Five
 - (4) Two
152. Find out the **wrongly** matched pair.
- (1) Squamous epithelium – Skin
 - (2) Ciliated epithelium – Bronchioles
 - (3) Glandular epithelium – Salivary gland
 - (4) Columnar epithelium – Body cavity
153. A tissue is observed to have cube-shaped cells with microvilli on their free surface. This tissue is found lining a tubular structure where it plays a major role in increasing the surface area for reabsorption. In which part of the body would you most likely find this tissue?
- (1) Inner lining of the fallopian tubes
 - (2) Walls of the air sacs (alveoli) in the lungs
 - (3) Proximal Convolute Tubule (PCT) of a nephron
 - (4) Lining of the buccal cavity
154. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The diameter of blood vessels, such as arteries and arterioles, can be altered to regulate blood flow and pressure.
Reason R: Spindle shaped striated muscle tissue is present in the walls of blood vessels with the ability to contract and relax.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both A and R are true, and R is the correct explanation of A.
 - (2) Both A and R are true, but R is not the correct explanation of A.
 - (3) A is true, but R is false.
 - (4) A is false, but R is true.

155. A genetic disorder prevents the formation of keratin in epithelial cells. Which of the following locations would be most severely affected?
- (1) The lining of the intestine, impairing absorption.
 - (2) The dry surface of the skin, compromising its protective function.
 - (3) The inner lining of blood vessels, affecting diffusion.
 - (4) The secretory ducts of salivary glands, blocking saliva release.

156. Read the following statements (A to D) and choose the option which states them as True (T) or False (F) **correctly**.

- Nerve cells do not form layer like epithelial tissues and remain structurally separate.
- Smooth muscles in a similar fashion to cardiac muscles always contract together e.g., large blood vessels.
- Epithelial tissues are highly vascular.
- Connective tissues are most abundant and widely distributed.

	A	B	C	D
(1)	F	F	F	F
(2)	T	T	F	F
(3)	F	F	F	T
(4)	T	F	F	T

157. Gap junctions are **not** found in;
- (1) Cardiac muscles
 - (2) Single-unit smooth muscles
 - (3) Epithelial tissue
 - (4) Differentiated skeletal muscle

158. A. All have collagen fibres.
B. All have fibroblasts.
C. All are connective tissues.
D. All are tendon.
E. All are ligament.
F. All are vascular in nature.
G. All are mesodermal in origin.

How many of the above given are related to areolar tissue, dense irregular connective tissue and dense regular connective tissue?

- (1) Seven
- (2) Five
- (3) Four
- (4) Two

159. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Heart	(I)	Travels along plasma membrane
(B)	Cockroach	(II)	Organ that consists of all four types of tissues
(C)	Frog	(III)	Represents invertebrates
(D)	Electrical disturbance generated in a neuron	(IV)	Represents vertebrates

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

160. Which of the following is **wrongly** matched pair?
- (1) Unicellular glandular cells – Goblet cells
 - (2) Cuboidal epithelium – Germinal epithelium of ovary
 - (3) Areolar tissue – Connects skin with muscle
 - (4) Eosinophils – Multilobed nucleus

161. Four healthy people in their twenties faced injuries resulting in damage and death of a few cells given below. Which of the cells are least likely to be replaced by new cells?

- (1) Liver cells
- (2) Neurons
- (3) Outer layer of the skin
- (4) Osteoblasts

162. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

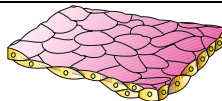
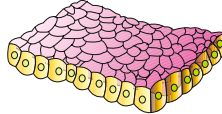
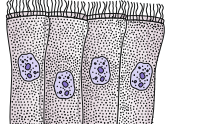
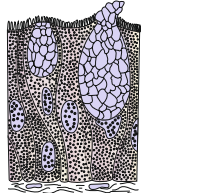
Assertion A: Adipose tissue stores excess circulating nutrients for long-term by converting them into triglycerides that accumulate within adipocytes.

Reason R: Adipose tissue is made mainly of densely packed chondrocytes that secrete fibrous collagen and calcium salts, forming a semi-rigid ground substance.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

163. Match **List-I** with **List-II**.

	List-I		List-II
(A)		(I)	Unicellular gland
(B)		(II)	Ciliated columnar epithelium
(C)		(III)	Pavement epithelium
(D)		(IV)	Cuboidal epithelium

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

164. Which of the following statement about cell junctions is **incorrect**?

- A. Gap junctions allow rapid transfer of ions and small molecules only.
- B. In almost all animal tissues specialised junction provide both structural and functional link between its individual cells.
- C. Tight junctions help to stop substances from leaking across a tissue.
- D. Adhering junctions provide cementing to keep neighbouring cells together.
- E. Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells.

- (1) B and C only
- (2) A and B only
- (3) Only A
- (4) A, B, C and D

165. Given below are two statements:

Statement I: In bone tissue, the hard matrix is arranged in concentric rings called lamellae.

Statement II: In cartilage, the pliable matrix is also arranged in concentric lamellae similar to that in bones.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Arrange from least rigid matrix to most rigid:

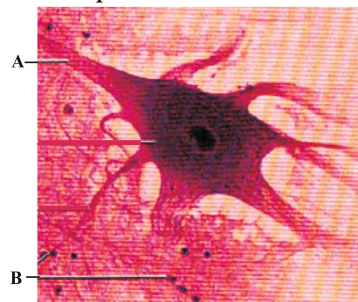
Blood (A), Areolar (B), Cartilage (C), Bone (D).

- (1) $D < C < B < A$.
- (2) $A < B < C < D$.
- (3) $A < C < B < D$.
- (4) $B < A < C < D$.

167. The progressive increase in the complexity of body organisation, from tissue level to organ level and then to organ-system level, as observed across the animal kingdom is a discernible pattern. This pattern is **best** described as an:

- (1) Morphological trend
- (2) Anatomical trend
- (3) Complexity trend
- (4) Evolutionary trend

168. Choose the option with **correct** labelling A and B.



	A	B
(1)	Axon	Neuroglia
(2)	Dendrite	Axon
(3)	Dendrite	Synaptic cleft
(4)	Axon terminal	Axon

169. Choose the **incorrect** statement about specialised connective tissues.

- (1) Chondrocytes occupy lacunae in a pliable matrix.
- (2) Osteocytes are embedded in a calcified, collagen-rich matrix.
- (3) In adults, the entire vertebral column is cartilaginous.
- (4) Bone marrow undergoes haemopoiesis.

170. Which of the following pairs **incorrectly** matches a specific anatomical location with the functional characteristics of the muscle tissue found there?

- (1) Wall of the urinary bladder: Uninucleate, non-striated fibres responsible for involuntary expulsion of urine.
- (2) Erector pili muscle of skin: Multinucleate, striated fibres responsible for voluntary piloerection ("goosebumps").
- (3) Biceps muscle: Multinucleate, striated fibres responsible for voluntary flexion of the forearm.
- (4) Wall of blood vessels: Uninucleate, fusiform fibres responsible for involuntary vasoconstriction and vasodilation.

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Hydra</i>	(I)	Simplest example of tissue level organisation
(B)	Contractility	(II)	Tissue chiefly responsible for peristaltic waves in gut
(C)	Elasticity	(III)	Ability to shorten on stimulation
(D)	Smooth muscle	(IV)	Ability to return to original length

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-I, B-II, C-III, D-IV

172. Why do basal cells of compound epithelium divide frequently?
- (1) To enhance nutrient absorption.
 - (2) To secrete hormones efficiently.
 - (3) To continually replace superficial cells lost to wear and tear.
 - (4) To increase mucus output.
173. Exocrine and endocrine glands are both derived from epithelial tissue. Which statement **correctly** distinguishes between them?
- (1) Both types of glands are unicellular only and are found scattered among other epithelial cells.
 - (2) Endocrine glands, like the salivary gland, produce enzymes, while exocrine glands produce hormones.
 - (3) Exocrine glands are ductless glands, whereas endocrine glands have ducts.
 - (4) Exocrine glands release their products through ducts, while endocrine glands release hormones directly into the fluid bathing the gland.
174. Given below are two statements:
Statement I: Blood is considered a specialized connective tissue because it connects different body systems, has an extracellular matrix (plasma), and originates from the mesoderm.
Statement II: The fibres in blood secreted by blood cells, such as fibrinogen, are always present in their active form, circulating in the plasma.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Statement I is correct but Statement II is incorrect.
 - (3) Statement I is incorrect but Statement II is correct.
 - (4) Both Statement I and Statement II are incorrect.
175. The sheath enclosing the several bundles of muscle fibres is _____ tissue.
- (1) epithelial (2) connective
 - (3) neural (4) both (1) and (2)
176. Read the following statements (A to D) and choose the option which states them as True (T) or False (F) **correctly**.
- A. A large variety of organisms are there in animal kingdom both unicellular and multicellular.
 - B. Connective tissue covers all four types of tissues.
 - C. A specialised type of connective tissue provides insulation to other tissues.
 - D. Skeletal tissues are named so as they are attached to bones of skeletal system.

	A	B	C	D
(1)	F	F	F	F
(2)	T	T	F	F
(3)	F	F	T	F
(4)	T	F	T	F

177. Which is **not** a function of muscle tissue?
- (1) Contract in response to stimuli
 - (2) Maintain postures
 - (3) Move body to adjust to changes in environment
 - (4) Play role only in few movements of the body
178. Given below are two statements:
Statement I: Camel's hump is composed of adipose tissue which provides water on oxidation.
Statement II: Dense irregular connective tissue is present in skin.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
179. A sample of connective tissue is characterized by a dense, interwoven network of collagen fibres and fibroblasts, providing structural strength in many directions. Where would this type of tissue most likely be found?
- (1) In a tendon, connecting muscle to bone.
 - (2) In the skin, just beneath the epidermis.
 - (3) As a covering on the ends of long bones within joint cavities.
 - (4) In the umbilical cord, as mucoid connective tissue.
180. While neurons are the primary units for impulse transmission, the majority of the neural tissue volume is composed of neuroglia. What is the main reason for the abundance of neuroglial cells?
- (1) They are the only cells in the neural system capable of mitosis.
 - (2) They are required to generate the electrical potential for neurons.
 - (3) They provide essential protection, structural support, and nourishment to the neurons.
 - (4) They are vestigial cells with no significant function in complex nervous systems.

SPACE FOR ROUGH WORK



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____